

Mock Exam: Data Management & Business Intelligence

90 multiple choice questions. Choose one answer (A-D) per question.

Business Intelligence

1. What does BI stand for?
 - A. Business Information
 - B. Business Intelligence
 - C. Binary Index
 - D. Budget Integration
2. What is the main purpose of BI?
 - A. Replace databases
 - B. Support decision making with data
 - C. Store raw logs only
 - D. Increase file system speed
3. BI is not an end in itself. It should be:
 - A. Isolated from business processes
 - B. Fully integrated into decision-making processes
 - C. Used only for compliance
 - D. Limited to IT departments
4. A data-driven culture means:
 - A. Decisions rely on data and analytics
 - B. Decisions rely on intuition only
 - C. Data stored in spreadsheets only
 - D. No data sharing allowed
5. Data silos are:
 - A. Shared company-wide data lakes
 - B. Isolated data systems across departments
 - C. Backups stored offsite
 - D. Encrypted databases
6. Shadow BI refers to:
 - A. Official BI tools used by IT
 - B. Unofficial BI practices outside central governance
 - C. Historical BI reports
 - D. BI training courses
7. Descriptive analytics answers:
 - A. What should we do?
 - B. Why did it happen?
 - C. What happened?
 - D. What will happen?
8. Operational BI (OBI) focuses on:
 - A. Long-term research only
 - B. Real-time or near-real-time operational decisions
 - C. Replacing data warehouses
 - D. Writing code without data
9. Business-data disconnect means:
 - A. Data is encrypted
 - B. Data insights are misaligned with business goals
 - C. Too many dashboards
 - D. Only IT can access data
10. Diagnostic analytics answers:
 - A. What happened?
 - B. Why did it happen?
 - C. What will happen?
 - D. What should we do?
11. The decision-action gap means:
 - A. Too much data
 - B. Decisions are made without data
 - C. Insights exist but are not acted on
 - D. Data is deleted too early
12. Predictive analytics answers:
 - A. What happened?
 - B. Why did it happen?
 - C. What will happen?
 - D. What should we do?
13. Organizational silos mainly refer to:
 - A. Too many dashboards
 - B. Lack of communication between departments
 - C. Too many backups
 - D. Slow internet
14. Prescriptive analytics answers:
 - A. What happened?
 - B. Why did it happen?
 - C. What will happen?
 - D. What should we do?
15. Netflix collects viewing data mainly to:
 - A. Reduce storage costs only
 - B. Understand engagement and guide content decisions
 - C. Replace all BI tools
 - D. Avoid dashboards

Python

1. Which is a valid Python variable name?
A. total_sales
B. 2nd_value
C. class
D. total-sales
2. What is the result of `len([1, 2, 3, 4])`?
A. 3
B. 4
C. 5
D. Error
3. Which type stores key-value pairs?
A. list
B. tuple
C. dict
D. set
4. What does this print? `for i in range(3): print(i)`
A. 1 2 3
B. 0 1 2
C. 0 1 2 3
D. 3 2 1
5. What is the result of `"data".upper()`?
A. "Data"
B. "DATA"
C. "data"
D. Error
6. What does `my_list.append(5)` do?
A. Adds 5 to the start
B. Adds 5 to the end
C. Removes 5
D. Sorts the list
7. What prints? `x = 5 if x > 3: print("yes") else: print("no")`
A. yes
B. no
C. 5
D. Error
8. What does this return? `def add(a, b): return a + b add(2, 3)`
A. 2
B. 5
C. 6
D. Error
9. For `items = [10, 20, 30]`, what is `items[0]`?
A. 10
B. 20
C. 30
D. Error
10. What does `import math` do?
A. Deletes a module
B. Imports the math module for use
C. Creates a new file
D. Runs a script
11. What is `sum([1, 2, 3])`?
A. 3
B. 5
C. 6
D. 7
12. What is `{"a": 1, "b": 2}["b"]`?
A. 1
B. 2
C. "b"
D. Error
13. What is `[x * 2 for x in [1, 2, 3]]`?
A. [2, 4, 6]
B. [1, 2, 3]
C. [2, 3, 4]
D. Error
14. What is `"a,b,c".split(",")`?
A. ["a,b,c"]
B. ["a", "b", "c", ""]
C. ["a", "b", "c"]
D. Error
15. For `items = ["a", "b", "c"]`, what is `items[-1]`?
A. "a"
B. "b"
C. "c"
D. Error

Database Architectures

1. What is a task of a file system?
 - A. Bring order to files/folders and metadata
 - B. Train ML models
 - C. Replace SQL
 - D. Encrypt all data
2. CRUD stands for:
 - A. Create, Read, Update, Delete
 - B. Copy, Run, Undo, Deploy
 - C. Cache, Restore, Update, Drop
 - D. Create, Replace, Use, Dump
3. The no free lunch theorem implies:
 - A. No backups are required
 - B. Performance tradeoffs exist (faster reads can mean slower writes)
 - C. Data is always consistent
 - D. Storage is free
4. In a row-oriented database, accessing by row index is typically:
 - A. Fast (direct access)
 - B. Slow because it scans all rows
 - C. Impossible
 - D. Only possible in the cloud
5. Searching by column value in a row-oriented DB is slow because:
 - A. Data is encrypted
 - B. You often scan rows to compare values
 - C. Rows are sorted by that column by default
 - D. Indexes are free and automatic
6. A transaction is:
 - A. A single query only
 - B. A unit of work treated as one
 - C. A backup file
 - D. A schema definition
7. An index is mainly used to:
 - A. Slow down reads
 - B. Speed up searches and lookups
 - C. Replace primary keys
 - D. Store images
8. Binary search complexity is:
 - A. $O(1)$
 - B. $O(n)$
 - C. $O(\log n)$
 - D. $O(n^2)$
9. A tradeoff of using indexes is:
 - A. No storage needed
 - B. Slower reads only
 - C. Extra storage and slower writes or updates
 - D. Data becomes raw
10. ACID stands for:
 - A. Atomicity, Consistency, Isolation, Durability
 - B. Accuracy, Control, Index, Data
 - C. Access, Compute, Integrate, Delete
 - D. Archive, Copy, Inspect, Deploy
11. A primary key:
 - A. Allows duplicate values
 - B. Uniquely identifies each row
 - C. Stores blobs
 - D. Is optional for tables
12. A foreign key:
 - A. Uniquely identifies rows in the same table
 - B. References a primary key in another table
 - C. Stores passwords
 - D. Replaces indexes
13. Document databases typically store:
 - A. Fixed rows and columns only
 - B. JSON-like documents with flexible schema
 - C. Only CSV files
 - D. Only images
14. SQL JOIN is used to:
 - A. Back up data
 - B. Combine rows from multiple tables
 - C. Delete a table
 - D. Change column types
15. Many-to-many relationships usually use:
 - A. A linking table
 - B. A single column only
 - C. A file folder
 - D. A view only

Data Management

1. A data warehouse is mainly:
 - A. Raw data storage without transformations
 - B. Central hub for cleaned structured data loaded via ETL
 - C. Only used for image storage
 - D. A file system replacement
2. ETL stands for:
 - A. Extract, Transform, Load
 - B. Encode, Transfer, List
 - C. Extract, Translate, Link
 - D. Export, Test, Log
3. A data lake typically:
 - A. Stores only structured data
 - B. Loads data without transformations
 - C. Requires fixed schema
 - D. Rejects text or audio
4. Staging tables are:
 - A. Final dashboard tables
 - B. Cleaned reusable tables for multiple marts
 - C. Only for backups
 - D. Raw unprocessed logs
5. A common risk of data lakes is:
 - A. Data becoming a data swamp due to lack of structure
 - B. Too much normalization
 - C. Only supporting SQL
 - D. Overly strict schemas
6. A mart table is:
 - A. A raw data dump
 - B. A dashboard-specific curated table
 - C. A file system index
 - D. A log archive
7. A data lake advantage is:
 - A. Low flexibility
 - B. Supports many data types and formats
 - C. Requires heavy transformations
 - D. Only for small datasets
8. A hybrid data lake:
 - A. Is a data warehouse only
 - B. Combines data lake and data warehouse but adds complexity
 - C. Removes the need for storage
 - D. Works only on-prem
9. On-premises vs cloud: which is correct?
 - A. On-prem uses provider data centers only
 - B. Cloud means local servers only
 - C. On-prem uses company-owned infrastructure; cloud uses hosted services
 - D. They are identical
10. A lakehouse:
 - A. Combines data warehouse and data lake with lower complexity
 - B. Is identical to a file system
 - C. Avoids query engines
 - D. Stores only logs
11. A good ETL process includes:
 - A. No documentation
 - B. Monitoring, testing, and lineage/traceability
 - C. Only manual steps
 - D. Deleting errors
12. Which is a common cloud service model?
 - A. KaaS
 - B. IaaS, PaaS, or SaaS
 - C. DaaS only
 - D. FaaS only
13. A high cost driver in a pure data warehouse can be:
 - A. Schema changes requiring large backfills
 - B. No storage
 - C. No transformations
 - D. Only text data
14. A data lake often needs:
 - A. No query tooling
 - B. Extra tooling for query and processing
 - C. Only spreadsheets
 - D. Only SQL views
15. A potential disadvantage of a lakehouse is:
 - A. It cannot store structured data
 - B. A standalone query engine may be needed
 - C. It blocks ETL
 - D. It only works on-prem

Introduction to AI

1. Artificial Intelligence is defined as:
 - A. Simulation of intelligence, especially learning
 - B. Only robots
 - C. Only databases
 - D. Only hardware
2. Era 1 (pre-deep learning) focused on:
 - A. Rules, logic, expert systems
 - B. Large-scale LLMs only
 - C. Purely random search
 - D. No data
3. Regression outputs:
 - A. A category label
 - B. A number
 - C. A document
 - D. An image
4. Supervised learning uses:
 - A. No labels
 - B. Labels for training
 - C. Only rules
 - D. Only random actions
5. Mean squared error (MSE) is:
 - A. A type of database
 - B. A common loss for regression
 - C. A clustering method
 - D. A storage format
6. Unsupervised learning example is:
 - A. Demand forecasting
 - B. Customer segmentation
 - C. Spam classification
 - D. Price prediction
7. Classification outputs:
 - A. A category label
 - B. A number
 - C. A file
 - D. A graph
8. Reinforcement learning learns by:
 - A. Labels only
 - B. Rewards and feedback
 - C. Fixed rules only
 - D. SQL queries
9. Gradient descent updates parameters by:
 - A. Moving in the direction of higher loss
 - B. Moving opposite the gradient
 - C. Random guessing only
 - D. No updates
10. k-means steps are:
 - A. Initialize, assign, update, repeat
 - B. Encode, train, deploy, monitor
 - C. Select, delete, migrate, backup
 - D. Sort, join, filter, group
11. k-means output is:
 - A. A single number
 - B. Cluster assignment for each point
 - C. A decision tree
 - D. A SQL query
12. Supervised learning example:
 - A. Customer churn prediction
 - B. Discovering clusters
 - C. Anomaly grouping without labels
 - D. Random guessing
13. Unsupervised learning uses:
 - A. Inputs with labels y
 - B. Only rewards
 - C. Inputs without labels
 - D. Only rules
14. The learning rate in gradient descent is:
 - A. A type of dataset
 - B. The step size for updates
 - C. A measure of accuracy
 - D. A storage limit
15. Training typically repeats until:
 - A. Loss stops improving
 - B. The dataset is deleted
 - C. Labels are removed
 - D. Queries fail

Generative AI and Agents

1. Generative AI primarily:
 - A. Creates synthetic data or content
 - B. Only stores data
 - C. Only classifies text
 - D. Only deletes data
2. An LLM mainly predicts:
 - A. Database schema
 - B. The next token
 - C. A SQL join
 - D. A file path
3. The context window is:
 - A. The model's training dataset size
 - B. The amount of text the model can use at once
 - C. The output length limit only
 - D. The GPU memory size
4. A hallucination is:
 - A. Perfect memory
 - B. Confident but incorrect output
 - C. A database index
 - D. A training label
5. A GAN has:
 - A. Encoder and decoder only
 - B. Generator and discriminator
 - C. Router and cache
 - D. Tokenizer only
6. Transformers use attention to:
 - A. Ignore all context
 - B. Capture dependencies and focus on relevant tokens
 - C. Remove tokens
 - D. Encrypt data
7. Encoder-decoder models are often used to:
 - A. Map input sequences to output sequences (e.g., translation)
 - B. Store images only
 - C. Run SQL
 - D. Compress files only
8. Vector databases exist mainly for:
 - A. Keyword search only
 - B. Semantic similarity search in embeddings
 - C. Storing backups
 - D. Running ETL
9. An embedding is:
 - A. A high-dimensional vector representation of data
 - B. A primary key
 - C. A file extension
 - D. A SQL query
10. RAG stands for:
 - A. Retrieve, Augment, Generate
 - B. Read, Analyze, Graph
 - C. Random, Approximate, Guess
 - D. Reduce, Aggregate, Group
11. A ReAct agent combines:
 - A. Reasoning and acting with observations
 - B. Only training
 - C. Only storage
 - D. Only encryption
12. A simple agent loop includes:
 - A. Goal, plan, tools, feedback loop
 - B. Tables, rows, columns, keys
 - C. Images, audio, video only
 - D. Compile, link, run, exit
13. Tool calls are:
 - A. External function interfaces the model can invoke
 - B. Only UI buttons
 - C. SQL backups
 - D. Static images
14. MCP stands for:
 - A. Model Context Protocol
 - B. Machine Code Parser
 - C. Multi Cloud Platform
 - D. Model Cache Process
15. ACP stands for:
 - A. Agent Communication Protocol
 - B. Automatic Code Pipeline
 - C. Access Control Policy
 - D. Agent Cloud Platform

Answer Key 1-B, 2-B, 3-B, 4-A, 5-B, 6-B, 7-C, 8-B, 9-B, 10-B, 11-C, 12-C, 13-B, 14-D, 15-B, 16-A, 17-B, 18-C, 19-B, 20-B, 21-B, 22-A, 23-B, 24-A, 25-B, 26-C, 27-B, 28-A, 29-C, 30-C, 31-A, 32-A, 33-B, 34-A, 35-B, 36-B, 37-B, 38-C, 39-C, 40-A, 41-B, 42-B, 43-B, 44-B, 45-A, 46-B, 47-A, 48-B, 49-B, 50-A, 51-B, 52-B, 53-B, 54-C, 55-A, 56-B, 57-B, 58-A, 59-B, 60-B, 61-A, 62-A, 63-B, 64-B, 65-B, 66-B, 67-A, 68-B, 69-B, 70-A, 71-B, 72-A, 73-C, 74-B, 75-A, 76-A, 77-B, 78-B, 79-B, 80-B, 81-B, 82-A, 83-B, 84-A, 85-A, 86-A, 87-A, 88-A, 89-A, 90-A